The Lorentz-Capacity Sub-Programme

Projective Origin of the Lorentz Factor in the Cosmochrony Programme

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Abstract

The Lorentz-capacity sub-programme answers a specific question in the Cosmochrony framework: where does the Lorentz factor γ come from, if spacetime geometry is not assumed as fundamental? The answer is that γ is not introduced as a kinematic postulate. It is reconstructed from the bounded Born–Infeld capacity budget of admissible projection, then lifted to the observer-transform structure of special relativity through the effective Lorentzian co-metric established in the geometric branch of the programme.

The completed chain is

$$\text{BI budget} \;\to\; F^\tau = \tfrac{1}{\gamma} \;\to\; g^{\mu\nu} = 2\eta^{\mu\nu} \;\to\; \Lambda(\beta)$$
$$\to\; R(x)^2 = 1 - \epsilon(x) \;\to\; \text{finite-}q \text{ domain.}$$

Six papers constitute the sub-programme. Paper [9] (LorCap) identifies the projective mobility parameter. Paper [10] (TempProj) constructs the temporal residual map $F_n^\tau = \sqrt{1 - B_n^2}$. Paper [4] (LCII) extends the construction to the fibrewise non-homogeneous regime. Paper [7] (LC-O1) reconstructs Lorentz transformations and length contraction. Paper [6] (LC-O2-O1) closes the local capacity–metric bridge without importing the Einstein equation. Paper [5] (LC-O3) identifies the exact finite-q domain of validity of the Carnot parabolic identification.

The result is not a new dynamical principle. It is a synthesis showing that time dilation, the light cone, Lorentz boosts, length contraction, local gravitational capacity lapse, and finite-q saturation corrections all follow from the same admissibility-capacity structure, once the Q5b–Q11 effective metric closure has been established.

Keywords. Lorentz factor; projective temporal ordering; Born–Infeld capacity; temporal residual map; Lorentz transformation; length contraction; capacity–metric bridge; gravitational time dilation; finite-q corrections; Cosmochrony.

Contents

1 Motivation **3**

2 The Structural Chain **3**

3 Paper-by-Paper Synthesis **5**
3.1 LorCap: mobility from projective capacity [9] 5
3.2 TempProj: the temporal residual map [10] 5
3.3 LCII: local fibrewise capacity lapse [4] 5
3.4 LC-O1: Lorentz transformations and length contraction [7] . . 6
3.5 LC-O2-O1: spectral closure of the capacity–metric bridge [6] . 6
3.6 LC-O3: finite-q domain of the Carnot identification [5] 7

4 Inputs and Outputs **7**
4.1 Upstream inputs 7
4.2 Downstream outputs 8

5 Status **8**

6 Conclusion **9**

1 Motivation

The Lorentz factor is usually introduced as part of the kinematics of special relativity, taken as foundational. In the Cosmochrony framework this is not satisfactory: the Lorentzian metric is itself an effective object reconstructed from admissible non-injective projection [2, 11]. The structural question is therefore: where does $\gamma = (1 - \beta^2)^{-1/2}$ come from?

The Lorentz-capacity sub-programme answers this by separating two roles that are fused in the standard presentation. First, the Born–Infeld capacity budget [3, 1] determines the admissible split between temporal and spatial projective capacity. Second, the Q5b–Q11 geometric closure supplies the effective Lorentzian interval on which observer transformations act as isometries.

The capacity relation

$$(F^\tau)^2 + \beta^2 = 1$$

is a Euclidean norm constraint in capacity space. It is not the Minkowski interval. Its role is to determine

$$F^\tau = \sqrt{1 - \beta^2} = \frac{1}{\gamma}.$$

The Lorentzian interval is supplied separately by

$$g^{\mu\nu} = 2\eta^{\mu\nu}.$$

The Lorentz group is therefore not the symmetry group of the capacity sphere. It is the isometry group of the effective Lorentzian metric. This distinction is the organising principle of the whole sub-programme.

The sub-programme also addresses two downstream questions. In the non-homogeneous regime, a local occupancy field $\epsilon(x)$ reduces the local residual temporal capacity; LC-O2-O1 closes the bridge $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$ spectrally, without any use of the Einstein equation. LC-O3 then establishes the exact finite-q domain in which the Carnot parabolic identification of the spatial load is valid.

2 The Structural Chain

Figure 1 displays the completed chain.

Each arrow has a distinct logical status.

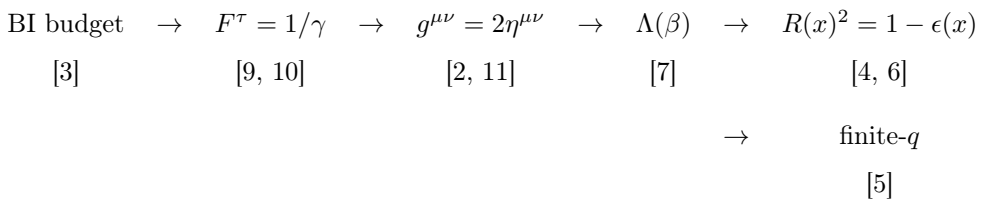


Figure 1: The Lorentz-capacity chain. Each arrow is closed by the paper(s) cited below it. The capacity sphere (left) and the effective metric (centre) are independent inputs; the Lorentz group acts on the metric, not on the sphere.

BI budget to temporal residual. The bounded capacity relation fixes the residual temporal capacity once the spatial load β is known:

$$F^\tau = \sqrt{1 - \beta^2}.$$

This is the projective origin of the Lorentz clock factor. The constraint comes from the Born–Infeld admissibility budget [3, 1]; its structural consequence for Lorentz mobility is established in [9].

Temporal residual to effective metric. The Q5b–Q11 closure [2, 11] supplies the homogeneous co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$, with no free parameter. The global factor 2 is irrelevant for the isometry group. This is the point at which Lorentzian geometry enters the chain.

Effective metric to Lorentz transformations. Once the metric is available, an inertial observer is a unit timelike splitting of the effective interval. The Lorentz boost $\Lambda(\beta)$ is the unique proper orthochronous isometry mapping one splitting to another at capacity load β, established in [7].

Homogeneous capacity to local lapse. The non-homogeneous extension introduces a local occupancy $\epsilon(x)$ and a residual capacity radius $R(x)^2 = 1 - \epsilon(x)$, defined in [4]. Paper [6] proves the equivalence $R(x)^2 = -2g^{\tau\tau}(x)$ from local admissibility data, closing the bridge spectrally.

Finite-q domain. The quadratic identity $B_n^2 + (F_n^\tau)^2 = 1$ is exact when B_n is the exact torus load. Paper [5] identifies the correction $\Delta_q(n) = 4m^2/q^2$ between the exact torus count and the Carnot parabolic approximation, and establishes the domain $n \leq R := (q-1)/2$ in which the approximation is valid.

3 Paper-by-Paper Synthesis

3.1 LorCap: mobility from projective capacity [9]

LorCap [9] establishes the first link between the admissibility capacity budget and relativistic mobility. The spatial load is encoded by a projective capacity parameter, and the temporal residual is interpreted as the clock-rate factor. The central result is the identification of a capacity variable whose Lorentzian reading is

$$F^{\tau} = \frac{1}{\gamma}.$$

LorCap also establishes why the naive total residual is insufficient: the central coordinate of the Heisenberg group contributes only a global phase, so the Lorentz factor must be extracted from the temporally projected residual rather than from the raw total capacity. The explicit construction of the temporal projection is completed by TempProj.

3.2 TempProj: the temporal residual map [10]

TempProj [10] constructs the temporal residual map as the unique positive function compatible with three inputs: the Q5b–Q11 Lorentzian splitting, the Born–Infeld unit capacity sphere, and the exact Weil filling law $\Sigma_n^{\mathrm{tot}} = 1 - B_n$. The result is

$$P_{\tau}^{Q5b}\,\Sigma_n^{\mathrm{tot}} = \sqrt{1 - B_n^2} = F_n^{\tau}.$$

This closes time dilation ($d\tau/dn = F_n^{\tau} = 1/\gamma$) and the light cone ($B_n = 1 \Rightarrow F_n^{\tau} = 0$). TempProj thereby proves that the Lorentz factor is the temporal residual of a bounded projective capacity budget. The reconstruction of the full transformation between inertial observers is the role of LC-O1.

3.3 LCII: local fibrewise capacity lapse [4]

LCII [4] extends the homogeneous construction to the non-homogeneous fibrewise regime. A local occupancy field $\epsilon(x)$ measures the fraction of projective capacity already locked in the fibre over x. The residual local capacity radius is

$$R(x)^2 := 1 - \epsilon(x).$$

The fibrewise temporal map factorises as

$$P_{\tau,x}^{Q5b}\,\Sigma^{\mathrm{tot}}(x,n) = R(x)\,\sqrt{f(x,n)\big(2 - f(x,n)\big)},$$

giving a capacity-language account of gravitational time dilation: local occupancy reduces the residual temporal capacity. LCII introduces two parallel definitions of $R(x)$,

$$R(x)^2 := 1 - \epsilon(x), \qquad R(x)^2 := -2g^{\tau\tau}(x),$$

whose analytic equivalence was left as an open bridge, closed by LC-O2-O1.

3.4 LC-O1: Lorentz transformations and length contraction [7]

LC-O1 [7] closes the observer-transform part of the sub-programme. The key conceptual clarification is that the Born–Infeld capacity sphere is not a Lorentzian structure: it supplies β and $1/\gamma$, while the Lorentzian metric is supplied by Q5b–Q11.

An inertial observer is a future-directed unit timelike vector u satisfying $\eta(u,u) = -1$, together with its simultaneity hyperplane $\Sigma_u = \{X \mid \eta(X,u) = 0\}$. The unique proper orthochronous isometry of η mapping the rest splitting $u_0 = \partial_\tau$ to the moving splitting

$$u_\beta = \gamma(\partial_\tau + \beta\,\partial_x), \qquad \gamma = (1-\beta^2)^{-1/2},$$

is the Lorentz boost

$$\tau' = \gamma(\tau - \beta x), \qquad x' = \gamma(x - \beta\tau).$$

Length contraction follows from measuring a rest interval on the moving simultaneity hyperplane:

$$\ell = \frac{\ell_0}{\gamma}.$$

Thus LC-O1 shows that projective temporal ordering does not merely produce a scalar clock factor. Once the effective Lorentzian metric is reconstructed, the same capacity load β fixes the complete inertial transformation.

3.5 LC-O2-O1: spectral closure of the capacity–metric bridge [6]

LC-O2-O1 [6] closes the non-homogeneous local bridge spectrally. The key definition is that $\epsilon(x)$ is not an arbitrary scalar density but a quadratic occupancy in the rank-three fibre sector $\mathrm{Sym}^2(V_\rho)$, normalised by the maximal Born–Infeld capacity $\sigma_{\max} = c_\chi^2$. The local admissibility-weight reduction lemma then gives

$$A_\tau(x) = 2\big(1 - \epsilon(x)\big),$$

and the principal-symbol identification in the LCII normalisation gives

$$g^{\tau\tau}(x) = -\frac{A_\tau(x)}{4}.$$

Combining these two identities yields the closure

$$R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x).$$

This is a local admissibility identity, not a gravitational field equation. No Einstein equation is used.

3.6 LC-O3: finite-q domain of the Carnot identification [5]

LC-O3 [5] resolves the saturation-boundary question. The quadratic capacity identity $B_n^2 + (F_n^\tau)^2 = 1$ is not corrected: it is exact whenever B_n is the exact finite-q torus load. The correction concerns only the pre-saturation Carnot parabolic identification $B_n^{\mathrm{parab}} = (2n^2 + 2n + 1)/q^2$.

Setting $R = (q-1)/2$ for odd prime q, the exact torus load satisfies

$$B_n^{\mathrm{torus}} = B_n^{\mathrm{parab}} \quad (n \le R), \qquad B_n^{\mathrm{torus}} = B_n^{\mathrm{parab}} - \frac{4m^2}{q^2} \quad (n = R + m).$$

The parabolic identification is asymptotically exact for $m/q \to 0$ and has a finite limit $\Delta_q \to 4(\alpha - 1/2)^2$ in the macroscopic folded regime $n/q \to \alpha > 1/2$. Thus LC-O3 establishes the domain in which continuum Carnot estimates are valid, without weakening the quadratic capacity law.

4 Inputs and Outputs

4.1 Upstream inputs

The sub-programme depends on three upstream results.

Born–Infeld admissibility budget [3, 1]. The local admissibility constraint gives the unit capacity form $(F^\tau)^2 + \beta^2 = 1$. Its role is to constrain admissible capacity splittings, not to supply a Lorentzian metric.

Projective temporal ordering [8]. The observable $I(n) = \sigma_{\mathrm{pair}}(0) - \sigma_{\mathrm{pair}}(n)$ supplies a non-decreasing temporal proxy. The sub-programme interprets $d\tau/dn = F^\tau$ as the effective clock-rate factor.

Q5b–Q11 metric closure [2, 11]. The geometric branch supplies the completed homogeneous co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$. This is the point at which Lorentzian geometry enters. The Lorentz group acts as the isometry group of this effective interval.

4.2 Downstream outputs

The sub-programme provides three outputs.

Special-relativistic kinematics. Time dilation ($d\tau/dn = 1/\gamma$), the light cone ($B = 1 \Rightarrow F^\tau = 0$), Lorentz boosts $\Lambda(\beta)$, and length contraction ($\ell = \ell_0/\gamma$) are all reconstructed from projective capacity data and the effective metric closure.

Local gravitational capacity lapse. The identity $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$ provides a capacity-side reading of gravitational time dilation, derived from admissibility-weight reduction before any use of the Einstein equation.

Finite-q control. The exact correction $\Delta_q(n) = 4m^2/q^2$ controls the domain in which continuum Carnot estimates of the spatial load can be trusted.

5 Status

Table 1 summarises the logical status of each paper.

Table 1: Logical status of the Lorentz-capacity sub-programme.

Label	Paper	Main result	Status
LorCap	[9]	Lorentz mobility from projective capacity	proved (cond. on $[U]$)
TempProj	[10]	$F_n^\tau = \sqrt{1 - B_n^2}$, time dilation	proved (cond. on $[U]$)
LCII	[4]	Fibrewise lapse $R(x)^2 = 1 - \epsilon(x)$	structural; bridge closed by LC-O2-O1
LC-O1	[7]	Lorentz boost; length contraction	proved
LC-O2-O1	[6]	capacity–metric bridge $R(x)^2 = -2g^{\tau\tau}(x)$	proved
LC-O3	[5]	Finite-q correction $\Delta_q = 4m^2/q^2$	proved

The sole remaining qualification is scope: the Lorentzian metric is inherited from Q5b–Q11, and the Born–Infeld capacity relation from the admissibility budget. No structural open problem remains inside the Lorentz-capacity sub-programme.

Remark (Scope of the residual). LC-O2-O1 proves the local identity $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$; it does not derive the Einstein field equations. The dynamical equation $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ belongs to the spectral entropy and gravity branch of the programme. The relation $\epsilon(x)$ to specific matter sectors

is a cross-branch question, not an internal open problem of the LC sub-programme.

6 Conclusion

The Lorentz-capacity sub-programme establishes the projective origin of the Lorentz factor in Cosmochrony. The essential result is the chain

$$\text{BI budget} \;\to\; F^\tau = \tfrac{1}{\gamma} \;\to\; g^{\mu\nu} = 2\eta^{\mu\nu} \;\to\; \Lambda(\beta)$$
$$\to\; R(x)^2 = 1 - \epsilon(x) \;\to\; \text{finite-}q \text{ domain.}$$

The Born–Infeld capacity sphere supplies the projective load β and the residual clock factor $1/\gamma$. The Q5b–Q11 geometric closure supplies the Lorentzian interval. LC-O1 reconstructs Lorentz boosts and length contraction as isometries of that interval. LC-O2-O1 closes the local capacity–metric bridge spectrally, without importing the Einstein equation. LC-O3 proves that finite-q corrections affect the Carnot parabolic identification, not the quadratic capacity law.

In this sense, the Lorentz factor is not postulated: it is the temporal residual of a bounded admissible projection. The Lorentz group is not imposed on the capacity sphere: it is the isometry group of the effective metric reconstructed from admissibility. The Lorentz-capacity sub-programme is therefore complete as a structural bridge between projective temporal ordering and relativistic kinematics.

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